E N G I N E E R I N G N O T E S

RadiNet Master Notes

Building the Flood-Monitoring System and the ASTE Dataset Paper

Project: RadiNet — Urban Flood Monitoring Benchmark, Ho Chi Minh City

Scope: Dataset curation, pipeline engineering, evaluation, submission

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Executive Summary

This document consolidates every decision, rule, and quantitative target agreed during the RadiNet project consultations into a single working reference. It is written for the two consumers of the same codebase: the big-data course project, which is graded on the streaming platform (ingestion, processing, storage, scalability, and demonstration), and the research paper, which is reviewed on validated dataset quality, benchmark rigor, and honest claims. The primary submission target is ICWSM 2027 Round 3 (dataset track), whose deadline of January 15, 2027 falls immediately after the end of the southern Vietnam flood season, making the season-long collection window the natural evaluation period.

The project enters this phase from a position of strength. The geographic knowledge base (122 officially published flood hotspots with five-decimal coordinates, 168 post-merger administrative units, OpenStreetMap cross-validation, and a self-learning gazetteer cache) is more complete than the infrastructure published in most comparable social-sensing papers. The annotated corpus already holds 3,922 sentences and 5,483 aspect-sentiment triplets, and raw data is organized into seven real flood episodes that form a ready-made evaluation harness. However, one defect is critical and must be fixed before any fine-tuning: the polarity distribution is NEG 94.2 percent, NEU 5.5 percent, POS 0.3 percent, which leaves roughly sixteen POSITIVE triplets in the entire dataset. This makes a trivial all-negative baseline reach 94 percent accuracy and removes the training signal required for the incident-resolution logic of the monitoring system.

The plan therefore proceeds in three phases: rebalance and re-govern the dataset (now to mid-November), verify and freeze it with human-verified gold test sets (mid-November to December), and build, benchmark, and write in parallel toward the January submission. Each section of this document states the rules to apply, the numbers to hit, and the defense scripts to use when the corresponding design decisions are challenged.

1. Situation Assessment: Two Deliverables, Current Assets, One Critical Defect

1.1 The dual-track principle

The course instructor and the paper reviewers evaluate different artifacts, and conflating them is the most common strategic error. The course grades the platform: whether the pipeline ingests continuously, processes a stream, stores in a scalable layout, demonstrates throughput and latency, scales horizontally with executors, and survives a replay demonstration without crashing. In that grading frame, the AI extractor is one replaceable

processing stage, and its F1 score is a variable rather than the criterion. The paper reviewers grade validated quality: per-event precision against ground truth, spatial recall against official and journalistic records, annotation credibility measured by inter-annotator agreement, and honest scoping of claims.

The two tracks share the same pipeline, and the architectural reason they can coexist is error averaging. Each post-level extraction is noisy, but an event is only created after at least three posts within sixty minutes, corroborated by two to three independent accounts on the same street, and passed through physical consistency checks (rain radar, tide above the 1.32 meter design threshold, and plausibility against the 122 officially listed hotspots). Aggregation at the event layer therefore suppresses per-post AI noise: the dashboard quality equals the event-layer quality, not the model-layer quality. This is also the honest answer to any concern that imperfect extraction accuracy will damage the course grade; what would actually damage it is a pipeline crash, a missing throughput story, or a dishonest claim.

1.2 Current asset inventory

The following inventory separates what is already strong from what still needs work, so that effort is spent only where the gaps are. The GIS stack and the episode organization are publication-grade; the class balance, the verification layer, and the evaluation numbers are not yet.

Table 1: Asset inventory and status

Asset group	Contents	Status and gap
GIS knowledge base	122 CSGT flood hotspots (83 rain, 39 tide) with 5-decimal coordinates; 168 administrative units per Resolution 1685; OSM centerline cross-check; self-learning gazetteer cache (110 records)	Strong. Keep as gazetteer source and soft prior, never as a hard gate (see 3.7)
Annotated corpus v2	3,922 sentences, 5,483 ASTE triplets, multimodal flags, episode and platform metadata	Scale adequate; polarity distribution defective; no human verification yet
Raw episode data	Seven episodes: Sep 10 rain, Sep 17 typhoon Soulik, Sep 27 tide, Oct 1 tide, Oct 7 rain, plus 2018-2025 archive and sporadic events	Strong. Forms the replay benchmark and evaluation harness
Real-time buffers	Latest TikTok and Facebook crawls, simulation pool, pipeline state with deduplication, crawl audit of 1,183 records	Good engineering hygiene; provides the funnel audit trail

Asset group	Contents	Status and gap
Multimodal assets	Extracted keyframes (one per two seconds), 16 kHz audio, sample videos	Sufficient for Tier 2 extension; not yet an annotated multimodal subset
Evaluation evidence	25 detected events, at least 10 distinct events per rain episode, Wilson CI 92 percent (75 to 98)	Feasibility-level only; needs rebalancing, verification, and per-episode validation

1.3 The critical defect: polarity imbalance

At 5,483 triplets, the recorded distribution of NEG 94.2 percent, NEU 5.5 percent, and POS 0.3 percent corresponds to roughly 5,166 negative, 301 neutral, and sixteen positive triplets. For the paper, this is a degenerate benchmark: a system that answers NEGATIVE for every triplet achieves 94 percent accuracy, macro-F1 becomes meaningless, and reviewers will reject the evaluation as trivially gameable. For the monitoring system, it is worse than a reporting problem: the incident-resolution feature, in which a POSITIVE triplet such as a report that the water has receded turns a red street green, has no training signal at all, so the deployed model would never produce the resolution output the pipeline design depends on.

The imbalance is a collection artifact rather than a property of reality. Crisis-focused crawl keywords over-sample complaint posts, and the annotating LLM may additionally default to NEGATIVE on any flood-adjacent sentence. Both mechanisms are correctable: positive posts demonstrably exist (drainage completion news, water-receding reports, praise of response teams), and the historical archive from 2018 to 2025 is a rich, untapped source of them. Section 2.4 specifies the mining plan, and Section 2.5 specifies the audit that estimates the false-NEGATIVE rate in the existing labels.

2. Dataset Curation Rules: Phase A, Now to Mid-November

2.1 Polarity governance

Polarity follows the SemEval-2014 Task 4 lineage: it expresses the attitude of the author toward the target aspect, not the physical presence of a flood. This single definition resolves the earlier ambiguity in which NEG was treated as equivalent to flooding. Downstream flood alerting is deliberately decoupled from polarity and driven instead by the aspect category and the opinion semantics, which is why a neutral factual measurement and an anguished complaint can both trigger the same incident through different routes.

Table 2: Polarity definitions with Vietnamese examples

Polarity	Definition	Examples
NEGATIVE	Complaint, criticism, distress, or grievance toward the aspect	ngap kinh khung (flood horrifically); xe chet may (vehicle stalls); nuoc tran vao nha (water pours into houses)
NEUTRAL	Factual statements, measurements, schedules, forecasts, and questions without evaluative emotion	ngap khoang 30cm (flooded about 30 cm); hoat dong tu 6h sang (operates from 6 a.m.); doan nay ngap chua? (is this section flooded yet?)
POSITIVE	Praise, satisfaction, relief, resolution, or effective response	nam nay het ngap (no more flooding this year); nuoc rut nhanh han (water recedes much faster); tram bom chay hieu qua (pump station runs effectively)

The annotator decision rule is one sentence and should be quoted verbatim in the annotation guidelines and the paper: first ask whether the sentence contains an explicit evaluation of the aspect; if yes, the sign of the evaluation decides POSITIVE or NEGATIVE; if no, the label is NEUTRAL regardless of whether flooding is mentioned. This rule removes most annotator disagreement in one step.

Two edge rules complete the routing. Interrogative posts never trigger incidents, because a question is a request for information rather than a report. A POSITIVE report counts as a resolution only when it is temporally and spatially close to an active incident; a dry-season statement that a street no longer floods is a status update, not a resolution event.

2.2 Aspect taxonomy: five classes

The taxonomy now carries five classes after two corrections from the earlier four-class draft: transit vehicles were removed because they are flood victims rather than drainage-domain entities, and the authority-response class was added because it is one of the dominant carriers of POSITIVE and NEGATIVE polarity in flood discourse. Each class has an explicit operational role in the pipeline so that no reviewer can ask why a non-triggering class exists in the dataset.

Table 3: Aspect taxonomy and pipeline roles

Class	Examples	Triggers incident?	Pipeline role
LOCATION_STREET	Tran Xuan Soan; Phan Huy Ich; ham chui An Phu	Yes (primary)	Red markers, spatial recall measurement

Class	Examples	Triggers incident?	Pipeline role
LOCATION_ADMIN	Phuong Linh Dong; TP Thu Duc; Quan 7	Yes	District-level aggregation and fallback localization
HYDROLOGY	Song Sai Gon; Kenh Nhieu Loc; trieu cuong (tide)	Yes (overflow)	River, canal, and tide context
DRAINAGE_INFRASTRUCTURE	cong (sewer); he thong thoat nuoc (drainage system); tram bom (pump station); nao vet (dredging)	No	Explanation layer; POSITIVE resolution corroboration; asset monitoring
AUTHORITY_RESPONSE	doi ung pho (response team); chinh quyen (authority); SCUFCP	No	POSITIVE and NEGATIVE carrier; response-quality evaluation

The boundary between HYDROLOGY and DRAINAGE_INFRASTRUCTURE is decided by one litmus test: would the entity still exist if humans stopped maintaining it? A canal would (HYDROLOGY); a pump station would not (DRAINAGE_INFRASTRUCTURE). Weather phenomena such as rain or tide may be annotated as aspects when genuinely evaluated, but they are excluded from alert triggering because they serve as the physical-consistency features of the validation layer. The infrastructure class exists because it is the drainage half of the dataset title, the main POSITIVE carrier, and the explanation layer; removing it would both break the domain claim and starve the POSITIVE class of hosts.

2.3 Span and schema rules (schema v3)

The single most important schema decision is that triplets keep verbatim surface spans exactly as written in the sentence. Canonicalizing the aspect text to gazetteer names would convert the task from span extraction to entity-linking ASTE and destroy comparability with the PASTE, STAGE, and EMC-GCN benchmark family. Categories and geographic identifiers instead become auxiliary annotation fields, which keeps the standard ASTE task intact and adds two bonus tracks (aspect category prediction and aspect linking) that increase rather than reduce the contribution.

Table 4: Schema v3 record structure

Field	Type	Rule
sentence	text	Raw sentence, diacritics as posted
triplets[].aspect_span	text	Verbatim substring of the sentence; head entity only, no prepositions or predicates
triplets[].opinion_span	text	Verbatim substring; must not overlap the aspect span
triplets[].polarity	enum	POSITIVE, NEUTRAL, or NEGATIVE per Table 2 decision rule
triplets[].aspect_category	enum	One of the five classes in Table 3
triplets[].severity	enum	Auxiliary severity label (none, mild, moderate, severe) for monitoring use
triplets[].gazetteer_id	string or null	Matched GIS entity when grounded
triplets[].grounding_provenance	enum	SPAN_MATCHED, FUZZY, or DOC_FALLBACK
metadata	object	platform (news, facebook, tiktok), episode_id, has_media, timestamp, post id

Span sanitation follows a belt-and-braces design. The head-entity rule is written into the LLM annotation prompt itself, so the error is prevented at generation time, and a regular-expression audit layer afterwards strips common prepositional prefixes (o, tai, trong, tren, vao, doan, khu vuc) and predicate leakage as a safety net. The audit must apply three guards to every triplet after any stripping: the aspect span must remain a substring of the sentence, the aspect and opinion spans must not overlap, and the span must remain unique and locatable. If stripping would produce an empty or predicate-only span, for example when khu vuc ngap loses its head noun, the original span is kept and the triplet is routed to document-level geolocation fallback instead. Every one of these checks belongs in the audit script so that the dataset ships with a machine-verified integrity report.

2.4 Rebalancing POS and NEU

The quantitative target is to add 800 to 1,000 POSITIVE triplets and 500 to 800 NEUTRAL triplets, bringing each class to at least 10 percent of the released dataset. If the target proves unreachable with real data, the honest fallback is to report the raw distribution, evaluate with per-class F1 and macro-F1, and optionally release a balanced split variant alongside the full data; fabricating or duplicating POSITIVE examples is not an acceptable option. Both the raw and released distributions belong in the paper so that the rebalancing is transparent.

Table 5: Mining sources for the rebalance

Class	Source	Method
POSITIVE	Historical archive 2018-2025	Drainage completion stories, dredging announcements, before-and-after coverage
POSITIVE	Targeted keyword crawl	hết ngập, nước rút, đỡ ngập, khô ráo, khen, hiệu quả and close variants
POSITIVE	Authority response discourse	Praise of response teams and timely intervention
NEUTRAL	News measurement reports	Water level figures, tide bulletins, official announcements
NEUTRAL	Operational statements	Pump schedules, station status, factual street reports

In parallel, audit the existing labels for annotation bias: draw a random sample of 200 records currently labeled NEGATIVE and have a human re-check them. If a non-trivial false-NEGATIVE rate appears, the LLM prompt is over-defaulting to negative and must be corrected with the decision rule and few-shot POSITIVE examples before the new labeling batch runs. This converts a suspicion about LLM bias into a measured number that can be reported.

2.5 Human verification protocol

The dataset credibility backbone is the separation of silver and gold labels. The training split remains silver: LLM-proposed labels with spot-checking, which is cheap and scalable. The development and test splits are gold: every record is human-verified, ideally by two to three annotators, with Cohen kappa reported at a target of at least 0.6 and an adjudication rule for disagreements. The LLM never decides a test label, because evaluating a model against labels produced by the same family of models is circular and reviewers correctly reject it.

The paper must also report the agreement between the LLM and the human verifiers on the audited sample, one short paragraph with percentages. This is the standard practice of the LLM-as-annotator literature: the LLM accelerates annotation roughly tenfold, humans remain the gold standard, and the agreement statistics make the acceleration visible and auditable. Gilardi and colleagues (PNAS, 2023) and the distillation literature provide the methodological cover for this design, so it should be presented as a deliberate method, not an apology.

2.6 Split strategy: episode-based, never random

Random splits leak: posts from the same flood episode share streets, vocabulary, and photographic evidence, so a random split places near-duplicates of test sentences into the training set and inflates all metrics. The split must therefore be temporal and episode-based.

The recommended allocation is train on episodes one to four plus the historical archive and sporadic events, development on episode five, and test on the October 7 episode, with the entire test split human-verified gold. Deviation is acceptable, but the principle is not: no episode contributes to both sides of any split boundary.

- Stratify the dev and test splits so that every polarity class has at least 30 instances; otherwise per-class F1 on the test set has no statistical meaning.
- Check v1 and v2 record overlap before freezing: any sentence present in both versions must appear in exactly one split.
- Record the split boundaries by episode identifier and timestamp range in the datasheet so that future users can reproduce the leakage-free evaluation.

2.7 Dataset definition of done

The following checklist is the acceptance gate for Phase A. No fine-tuning run should consume the dataset before every box can be ticked, because each unchecked box is either a degenerate training signal or a reviewer rejection risk.

Table 6: Dataset acceptance checklist

Done	Checklist item
□	POS at least 10 percent and NEU at least 10 percent of released triplets
□	Every triplet: verbatim aspect span is a substring of its sentence
□	Every triplet: aspect span and opinion span do not overlap
□	Every triplet: aspect category assigned from the five-class taxonomy
□	Every grounded triplet carries gazetteer id and provenance flag
□	Test split is 100 percent human-verified gold, episode-based, no leakage
□	Cohen kappa and LLM-human agreement reported for verification sample
□	Vehicles removed from aspects; degenerate spans routed to fallback
□	Severity labels present as auxiliary field
□	Datasheet, ethics statement, license, and release plan drafted

3. System Build Rules: Phase B, October to December

The pipeline architecture decided in the earlier design rounds remains valid and requires only four node-level updates to absorb everything learned since. This section records the architecture, the updates, and the concrete engineering checklist for each stage, in the order the stages appear in the flow. The build runs in parallel with Phase A because stages three

to six are prompt and code work rather than training work; only the fine-tuned model itself waits for the frozen v3 dataset.

3.1 Pipeline architecture and the four diagram updates

The six-stage flow runs from social sources (Facebook, TikTok, news sites) through polling collectors on a 15-to-30-minute cycle into Kafka, then through modality extraction (title text with emoji, PhoWhisper audio transcription, CLIP-selected frames), the AI extraction service with a fast path and a slow path plus the fallback chain, the validation and aggregation stage, and finally the serving layer of PostGIS, Elasticsearch, the live dashboard, and the human review hold queue. Figure 1 shows the reference diagram, which is also available as a Mermaid source file for the paper.

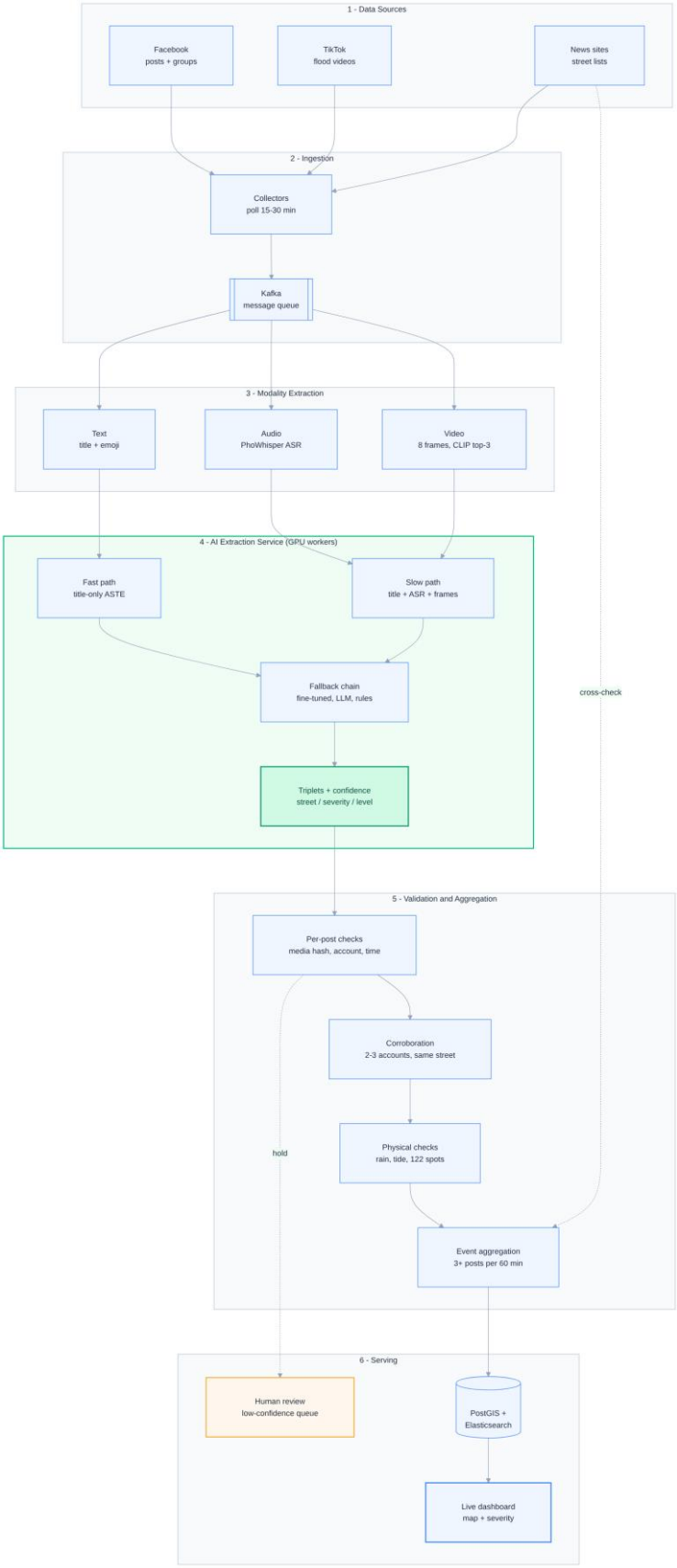


Figure 1: RadiNet AI extraction pipeline — six stages from data sources to serving, with the news cross-check as a dashed verification edge.

Table 7: The four diagram v2 updates

Node	Before	After	Reason
Triplet output	street / severity / level	aspect + opinion + polarity + severity + confidence	Polarity is annotated and alerting is decoupled from it; routing happens downstream
New geo-resolution stage	not shown	diacritic-normalized fuzzy gazetteer matching with provenance flags	The GIS stack is the system backbone and must be visible; provenance statistics are a paper table
Event aggregation	create at 3+ posts per 60 min	full lifecycle: create, resolve, close	The POSITIVE class exists to turn streets from red to green; resolution needs a path
Fallback chain	fine-tuned, LLM, rules	same, annotated as trained on dataset v3	The deployed model quality is bounded by the rebalanced dataset

3.2 The fine-tuned extraction model

For the fast path, an encoder-based span or table model on PhoBERT-base is the right choice: millisecond latency, single-GPU serving, batch-friendly inference, and no dependency on Vietnamese syntactic parsing, which rules out the dependency-tree-heavy ASTE architectures. The generative ASTE approach already planned in the repository is a good fit for the slow path and the benchmark comparison, where its flexibility justifies the decoding cost. The model must never be trained on the current 94/5.5/0.3 distribution, because that would bake the degeneracy into the production system; training consumes the frozen v3 dataset only.

Table 8: Training and serving checklist

Done	Checklist item
□	Base encoder PhoBERT-base with span or table heads (boundary-driven table-filling family)
□	Diacritic-dropout augmentation on 20 to 30 percent of training sentences (TikTok and Facebook text is frequently diacritic-less)
□	Emoji retained as tokens, not stripped, because they carry sentiment signal in flood posts
□	Metrics: exact-match triplet F1, per-polarity F1, macro-F1, and span F1 on the gold test split
□	Confidence calibration by temperature scaling so the fallback-chain escalation threshold is meaningful

Done	Checklist item
□	ONNX export with batch inference; latency budget at most 50 milliseconds per post at p95
□	Training turnaround kept under one hour per run on a single GPU to allow cheap iteration

3.3 The decision layer: routing triplets to actions

The governance specification routes triplets to pipeline actions through explicit rules rather than through polarity alone, and these rules should be implemented as one isolated, testable module between the extractor and the validation stage. Keeping them in code with unit tests is what makes the alerting behavior auditable, which matters both for the course demonstration and for the paper's system description.

Table 9: Routing rules

Condition	Action
LOCATION or HYDROLOGY aspect + flood opinion + NEGATIVE or NEUTRAL + assertive sentence	Incident signal
POSITIVE + resolution vocabulary + temporal or spatial proximity to an active incident	Resolution signal (red to green)
Interrogative sentence	No trigger; log as information request
Weather aspect (rain, tide) evaluated	Context record only; never a trigger
Model confidence below threshold	Escalate through the fallback chain
DRAINAGE_INFRASTRUCTURE or AUTHORITY_RESPONSE aspect	Explanation and asset-status records; resolution corroboration when positive

3.4 Geo-resolution service and provenance

Grounding converts an extracted aspect span into a coordinate and a persistent GIS entity. The resolution order is: normalize diacritics, fuzzy-match with a library such as rapidfuzz against the 122 hotspot file, the 168 administrative unit file, and the dynamic cache; on a miss, query OpenStreetMap and write the result into the dynamic cache so it is never queried again; finally emit the provenance flag. The three flags are SPAN_MATCHED, FUZZY, and DOC_FALLBACK, and the proportion of aspects in each bucket is itself a publishable statistic that quantifies dependence on the official list.

The validation stage adds a grounding gate: only SPAN_MATCHED or high-confidence FUZZY aspects may trigger incidents, while DOC_FALLBACK aspects route to the hold queue for human review. This single gate removes an entire class of false positives in which a document-level location guess would otherwise create a map marker from a weak signal.

3.5 Event lifecycle rules

An event is created when at least three posts within sixty minutes agree on the same street or district from at least two independent accounts, with the consolidation window set to four hours for rain-driven events and twelve hours for tide-driven events. An event is resolved when a POSITIVE resolution signal arrives near the active incident, and closed after two hours of silence with a cooldown of 60 to 90 minutes to prevent resurrection of the same episode. Every event records its alert latency, measured from the first post timestamp to the trigger moment, because that latency is the retrospective performance metric for the paper and the demonstration.

These defaults are starting values, not eternal truths. The window-selection methodology from the earlier analysis applies: profile the burst anatomy of historical episodes (first post, peak, and the 90-percent point), sweep candidate trigger and consolidation windows against the ground-truth street lists, select the window that satisfies the objective, and validate on a held-out episode. The final window table belongs in the paper appendix.

3.6 LLM fallback integration

The LLM serves two bounded roles and nothing else. Offline, it is the annotator that proposes dataset labels under human verification. Online, it is the escalation tier for low-confidence extractions, using exactly the same annotation prompt rules as the dataset labeling: head-entity spans, the polarity decision rule, and the five-class categories. Reusing the same prompt is what keeps the training data and the escalation output semantically consistent, so the fine-tuned model and the LLM agree on the task definition rather than silently diverging.

The integration needs a per-day API cost cap, batched escalation rather than per-post calls, and the discipline that escalated outputs still pass the grounding gate and the per-post checks. The LLM is never the hot path, for four reasons that also form the defense script: a frozen model is reproducible while an API drifts and deprecates; local inference has millisecond latency and no rate limits; the marginal cost per post is near zero; and citizen data does not leave the country. Calling an API from the streaming layer demonstrates no platform skill, whereas serving a fine-tuned model demonstrates the full stack, which is what the course grades.

3.7 The 122-hotspot principle: soft prior, never a gate

The official flood hotspot list is a strength that becomes a rejection risk only through one specific misuse: a hard gate that accepts events only at listed streets. A gate would make precision true by construction rather than by measurement, would make the gap-filling contribution structurally impossible, and would enforce the stale priors of a static snapshot,

since the list still contains streets that have been fixed and misses streets that newly flood. The list instead plays three legitimate roles: one gazetteer source among several (the dynamic cache and OSM resolve everything else), a soft confidence prior in the physical checks, and one of three evaluation references alongside the IRD Dataverse archive and per-episode news street lists.

Table 10: Five defensive moves against the list-dependence critique

Done	Checklist item
□	Report grounding-provenance statistics quantifying how much of the grounding uses the list
□	Run the with-list and without-list ablation and report both rows
□	Report N cross-verified events at streets absent from the official list (the undocumented-detection headline)
□	Report directional disagreement: listed streets never observed or observed as resolved (stale entries)
□	Cite the list as a dated, versioned snapshot (October 2026 publication) and note the periodic revisions

4. Evaluation and Big-Data Demonstration Rules

4.1 How big-data claims are proven

The governing principle is that event count is an output metric, the Value of the 5V framework, while big-data claims in the published literature are always justified on the input side: crawl volume, monitoring duration, latency and throughput, and the funnel compression ratio. Twenty-five events answer the question of whether the system works, not whether it is big. The demonstration therefore reports the funnel from raw posts to events, the posts-per-day rate, the platform split, and the replay throughput, with the N-to-25 compression ratio as the headline figure, in direct analogy to the published systems in Table 11.

For the course demonstration, the seven-step replay script remains the plan: accelerated replay of a recorded episode, live throughput panel, latency percentiles, the funnel panel, horizontal scaling by adding executors, a fault-tolerance kill-and-recover segment, and the validated October 6-7 episode as the showcase. This demonstrates every platform skill the course grades while requiring no live weather.

4.2 Anchor figures from related work

The following table is the evidence base for the scale argument and the citation anchors for the paper. Note the CrisisMMD precedent in the last row: a small corpus with no big-data claim at all is still a top-cited dataset paper, which is the escape hatch if the input denominator stays modest.

Table 11: How published systems justify scale

Paper	Input scale	Output and performance
de Bruijn et al. 2019, Scientific Data	88M tweets, 4 years, 12 languages	10,000+ flood events, about 90 percent precision, 63 percent recall versus NatCatSERVICE
Gu et al. 2026, Nature	92.98M Sina Weibo posts plus GSMaP satellite precipitation	Flood footprint tracing; detects previously undocumented events
Kryvasheyeu et al. 2015, PNAS	50M+ tweets for a single hurricane (Sandy)	Damage proxy with 0.55-0.81 correlation
Huang et al. 2022, EDEE (operational, China)	Continuous multi-source stream	3,170 events in 6 months, 93.56 percent clustering accuracy, 12.46 percent of alerts within 1 hour
Sakaki et al. 2010, WWW	Real-time Twitter stream	96 percent of earthquakes at JMA intensity 3 or above
Earle et al. 2011, USGS	Twitter stream	48 earthquakes in 5 months with 2 false triggers
PetaJakarta	One city, one monsoon	1,119 confirmed flood reports
CrisisMMD, AAAI 2018	About 41K tweets and 2K images	Multimodal labels; small corpus, no big-data claim, still highly cited

4.3 The 5V checklist

Each 5V dimension is mapped to the specific number that proves it. The Volume row is the currently missing denominator and is the single most important statistic to start recording immediately, because it cannot be reconstructed retroactively if the crawl logs are not kept.

Table 12: 5V status and required evidence

Dimension	Required evidence	Status
Volume	Total posts N, posts per day, platform split, raw corpus size	Missing denominator; record from now on
Velocity	Streaming path (Kafka, Flink or Spark), posts per second in replay, per-post latency percentiles	Demonstrable via replay benchmark

Dimension	Required evidence	Status
Variety	Facebook plus TikTok, text plus ASR audio plus frames	Rare strength; strongest V
Veracity	Five-layer validation, LLM labeling with human verification, kappa	Partial until verification protocol runs
Value	25 events, at least 10 per episode, Wilson CI 92 percent (75-98)	Feasibility level; grows with the season

4.4 Episode replay benchmark harness

The evaluation harness replays the seven recorded episodes through the pipeline at accelerated speed, which turns the historical archive into a controlled benchmark with known ground truth. For each replayed episode, measure per-event precision against the news street lists, spatial recall against the union of news lists, the IRD Dataverse archive, and the official 122-spot list, simulated alert latency as the trigger time minus the first post timestamp, and throughput as a function of the number of GPU workers to produce the scalability curve. The 100-post regex-versus-model comparison harness (missed streets, false matches, severity assigned) runs on the same episodes and becomes the baseline row of the benchmark table, converting the teacher challenge into a result.

Spatially, the target numbers to report are the per-episode street-level recall (ten detected against roughly 12 to 40 news-reported streets per rain), district coverage, and the count of detected locations outside the official list, each cross-verified. These are exactly the numbers the IRD Dataverse archive (425 geolocated observations, 2002-2026, DOI 10.23708/8Y16HU) was identified to validate.

4.5 Model benchmark table design

The benchmark table is the paper's core empirical artifact and must include the LLM rows so that the strongest objection is answered inside the paper rather than in the rebuttal. The literature expectation is that LLMs are strong at sentence-level classification but weak at token-span structured extraction, especially in Vietnamese with dropped diacritics, so the comparison itself is a finding whichever way it resolves.

Table 13: Benchmark table design

System	Triplet F1	Per-class F1	Latency per post	Cost per 1K posts
Regex and rule baseline	to measure	to measure	sub-ms	negligible
PhoBERT fine-tuned (fast path)	to measure	to measure	about 50 ms p95	GPU amortized

System	Triplet F1	Per-class F1	Latency per post	Cost per 1K posts
PASTE / STAGE / EMC-GCN style	to measure	to measure	to measure	GPU amortized
GPT or Gemini, zero-shot	to measure	to measure	seconds, rate-limited	per-call API
GPT or Gemini, few-shot	to measure	to measure	seconds, rate-limited	per-call API

Fill the measurement columns after the frozen v3 test split exists; the design point is that LLM rows are baselines in the comparison, not competitors to the deployed system.

4.6 Event validation against ground truth

Event-level validation triangulates three references with different biases: the IRD Dataverse archive as the long-run scientific record, the per-episode news street lists as the journalistic record of each rain, and the official 122-spot list as the government planning prior. Reporting agreement and disagreement with all three, rather than optimizing against any single one, is what makes the evaluation credible, and the disagreements are often the interesting result: detections outside all three references, when corroborated by two or three independent accounts, are the previously undocumented flood points that mirror the Nature-paper finding.

The supply-side argument completes the feasibility story for any reviewer or examiner who doubts that citizens post enough about flooding. Vietnam has 76.2 million active social media identities, roughly 75 percent of the population, with Facebook reach near 77 percent and TikTok near 41 percent, so the sensor network theoretically exists; the empirical observation that every officially confirmed flooding episode yields at least ten distinct street-level events across the city demonstrates the supply in practice. The episode-level distribution table with minimum, median, and maximum events per episode, and the margin between the trigger threshold of three and the observed minimum of ten, is the rigorous form of that argument.

5. Paper Writing Rules: Phase C, December to January

5.1 Retrospective evaluation framing

The season ends after the system does, and that is fine, because the collection pattern has a name in the literature: prospective collection with retrospective processing. Posts are timestamped, so a lightweight collector running during the season preserves everything the finished pipeline needs, and the evaluation then replays the recorded stream. de Bruijn's system runs historic and real-time dual modes, CrisisMMD was collected after the disasters,

and the FEDA system validated retrospectively against EM-DAT, so reviewers accept this pattern readily for dataset papers. What the pattern buys is the full claim set of a retrospective evaluation; what it costs is the claim to live operation, which moves to future work.

Table 14: Claimable and not claimable under retrospective evaluation

Claimable	Not claimable
Per-event precision against ground truth	Live operational uptime
Spatial recall per episode against news and archive records	Real alert delivery to end users
Simulated alert latency (trigger minus first post timestamp)	Continuous availability during the season
Funnel statistics and throughput from replay	True production cost accounting
Flood-level accuracy from the episode records	Unattended long-run stability

The limitations paragraph should state the boundary in one sentence: the system was evaluated by retrospective replay of prospectively collected data covering the 2026 season, and a prospective live deployment is planned for the 2027 season as future work. Honest scoping of this kind strengthens rather than weakens the paper, because the alternative, an implied live claim the logs cannot support, is what reviewers punish.

5.2 Paper skeleton: nine sections

The dataset-paper skeleton below is the agreed structure, and it maps one-to-one onto the material in this document, so writing in December is assembly rather than invention. Each section lists its source material so the drafting order can follow the availability of the frozen numbers rather than the narrative order.

Table 15: Paper skeleton and source material

Section	Content	Source
1. Introduction	HCMC flooding problem, sparse fixed sensors versus the 76M-user social sensor network, absence of any Vietnamese flood ASTE dataset	Sections 1 and 4.6 of this document
2. Related work	ASTE benchmark lineage; crisis social media (CrisisMMD, de Bruijn, DisasterMM); Vietnamese ABSA (VLSP 2018)	Appendix B anchors
3. Data collection	Crawl funnel table, platform split, season coverage, episode organization	Funnel table (4.1) and episode archive

Section	Content	Source
4. Annotation	Polarity decision rule, five-class taxonomy, silver-gold protocol, agreement statistics	Tables 2, 3, and the verification protocol (2.5)
5. Dataset analysis	Class distribution, severity distribution, triplet density, span statistics	Frozen v3 metrics
6. Benchmark	Comparison table with regex, fine-tuned, architecture, and LLM rows; per-class F1	Table 13 design
7. Event validation	Per-episode precision, spatial recall triangulation, simulated alert latency, undocumented detections	Sections 4.4-4.7
8. Ethics and limitations	Public data, anonymization, platform terms, retrospective scope	Section 5.4
9. Conclusion	Contributions: first Vietnamese flood ASTE benchmark, grounding bonus tracks, gap-filling finding	Executive summary

5.3 Figures and tables inventory

The paper needs seven visual artifacts, and all of them can be prepared before the benchmark numbers freeze, which allows the writing to proceed in parallel with the final experiments. The pipeline diagram already exists in rendered form and as Mermaid source; the funnel and benchmark tables are templates awaiting numbers; the class-distribution chart and latency distribution are single matplotlib figures from the frozen dataset and replay logs; and the spatial-recall table is a direct export from the validation harness.

- Figure 1: the six-stage pipeline diagram (already rendered, reused in Section 3 of this document).
- Table: crawl-to-event funnel with posts per day and platform split.
- Figure: polarity class distribution, raw versus released.
- Table: benchmark comparison (Table 13 design) with per-class F1.
- Table: spatial recall per episode against the three references.
- Figure: simulated alert latency distribution across episodes.
- Figure: burst anatomy of a representative episode (first post, peak, 90 percent point).

5.4 Datasheet, ethics, licensing, and release

A dataset paper is judged substantially on its documentation, so the release package is part of the contribution rather than an afterthought. The datasheet records the collection protocol,

the funnel, the platform split, the episode coverage table, the class statistics, the annotation methodology with agreement numbers, and the split boundaries by episode identifier. The ethics statement covers the use of public data only, the hashing of user identifiers during release, the platform terms of service, and the crisis-context sensitivity of the content. The license should be a research-standard choice such as CC BY-NC-SA, and the release should go to GitHub and HuggingFace with a versioned Zenodo DOI so the dataset is citable as an artifact.

5.5 Multimodal scoping: Tier 1 and Tier 2

The multimodal ambition is real but must be scoped so that the January deadline is not hostage to it. Tier 1, the paper core, is text ASTE over the title plus the PhoWhisper transcript, which already exercises the audio modality through the ASR merge. Tier 2, the extension, is the frame subset: keyframes selected by the CLIP ranking with CrisisMMD-style severity labels, either as a released subset with statistics or as a clearly labeled future-work section. The paper must not claim a multimodal benchmark unless the frame subset is fully annotated, and the PhoWhisper transcription quality needs a word error rate audit on a sample before its output is treated as ground text.

6. Submission Strategy and Defense Preparation

6.1 Venue ladder

The ladder below reflects the verified deadlines as checked during the earlier planning round. The primary target is ICWSM Round 3 because dataset papers are accepted in the January round only, retrospective evaluation is well precedented there, and the timing matches the end of the season. The KDD Datasets and Benchmarks track in February is the comfortable backup that additionally values the funnel and scale story. Deadlines that closed in October 2026 (NAACL and COLING via ARR, WWW) are excluded; the ARR capacity cap from the October 2026 cycle onward makes later ARR cycles slower to plan around.

Table 16: Venue ladder with deadlines

Venue	Deadline	Role
ICWSM 2027 Round 3, Edinburgh	January 15, 2027	Primary target; dataset track, January round only
KDD 2027 Datasets and Benchmarks, cycle 2	about February 2027	Backup; values the funnel and scale evidence
IJCAI 2027 (abstract)	January 12, 2027	Tight; research track, less dataset-oriented
SIGIR 2027 full papers	January 21, 2027	Option if the benchmark story strengthens

Venue	Deadline	Role
ARR cycles to ACL or NAACL	rolling, 2027	Later option; capacity-capped since October 2026

The strategic split remains: submit the dataset paper in January, and keep a second, system-level paper for a prospective 2027 season with live deployment, aimed at a venue that values the big-data platform story. Trying to force both contributions into one submission weakens each of them.

6.2 Master timeline

Table 17: Master timeline to submission

Window	Work
Now to mid-November	Keep the collector running; archive per-rain news street lists; mine POSITIVE and NEUTRAL sources; implement schema v3 with prompt rules and the audit script; validate the October 6-7 episode
Mid-November to December	Human verification of dev and test splits; kappa and agreement statistics; episode-based splits; freeze the dataset
December	Fine-tune the fast-path model on v3; wire the decision layer, geo-resolution, lifecycle, and LLM fallback in parallel
Late December	Run the benchmark table and the episode replay evaluation; produce all figures
January	Write and assemble the paper; submit ICWSM Round 3 by January 15
February	KDD Datasets and Benchmarks backup if January is missed or reviews warrant resubmission

6.3 Defense scripts for the four anticipated challenges

Challenge 1: prove the big-data aspect

Script: the claim axis is the input side, not the event count. We report the crawl-to-event funnel, posts per day, platform split, replay throughput and latency, and the horizontal scaling curve, in the same format as the published systems (88M tweets in de Bruijn, 92.98M posts in the Nature paper, 50M for one hurricane). The event count is the Value output of that funnel, exactly as in those systems. CrisisMMD, with 41K tweets, shows that a dataset paper does not need a big-data claim at all, so the claim is calibrated to the denominator we can actually document.

Challenge 2: why not use the LLM directly instead of the ASTE model

Script: the LLM was my annotation tool, offline, one-time, and human-verified on the gold sets; it never runs in the pipeline. The AI in the pipeline is the fine-tuned extractor, because production needs a frozen model that scales with the platform: millisecond latency, no API dependency, no per-post cost, full reproducibility. The benchmark shows how both compare on this task; the LLM rows are baselines in the comparison, not competitors to my system. Training a deployable student model on LLM-labeled data is knowledge distillation, the methodology of Gilardi (PNAS 2023) and the ZeroGen line, not a shortcut.

Challenge 3: the 122-list dependence critique

Script: the official list is one gazetteer source and a soft plausibility prior; detection is list-independent by design, because the dynamic cache and OpenStreetMap resolve any street, which we quantify with grounding-provenance statistics and a with-and-without-list ablation. It is also only one of three evaluation references, alongside the IRD Dataverse archive and the per-episode news street lists. Notably, the system detected N cross-verified flood points absent from the official list; that gap-filling is the contribution.

Challenge 4: signal sufficiency, windows, and regex

Script: sufficiency is the founding question of social sensing, and the answer has two sides. Theoretically, 76 million Vietnamese social media users constitute the sensor network (the citizens-as-sensors framing, DataReportal supply numbers). Empirically, every officially confirmed episode yielded at least ten distinct street-level events against a trigger threshold of three, a margin reported as an episode-level distribution table. The detection window is a measured quantity, profiled from burst anatomy and validated by backtest sweep, not a guess. And regex is the floor, not the ceiling: it is the baseline row of the benchmark and the last tier of the fallback chain, while the AI supplies severity, paraphrase coverage, and map-ready geolocation.

6.4 Risk register

Table 18: Risk register

Risk	Mitigation
POSITIVE mining falls short of target	Expand keyword set, mine the archive deeper, and fall back to reporting the raw distribution with per-class F1 and a balanced split variant
Timeline overruns past January 15	KDD Datasets and Benchmarks in February; the retrospective patterns keep every claim valid regardless of submission date
LLM escalation cost surprises	Per-day cost cap, batched escalation, and the rules tier as the floor

Risk	Mitigation
API or model drift breaks reproducibility	The deployed model is frozen; the LLM appears only in bounded escalation with logged versions
Social data decays before collection ends	Collect-now urgency: Facebook group search depth and TikTok feed recency are limited, so archive continuously
Reviewer suspects evaluation circularity	Ablation rows, provenance statistics, and the human-verified gold test split are pre-emptive answers

Appendix A: Consolidated Master Checklist

This is the single list to work from. It merges the dataset gate (Section 2.7), the system build items (Section 3), the evaluation items (Section 4), and the paper and submission items (Sections 5 and 6) into one tick-box sequence ordered by dependency: dataset first, system in parallel, evaluation on the frozen dataset, then paper and submission.

Table 19: Master checklist

Done	Checklist item
□	Schema v3 implemented with verbatim spans and auxiliary fields
□	LLM annotation prompt carries head-entity, polarity, and category rules
□	Audit script validates spans, overlap, categories, dedup, and v1/v2 overlap
□	POSITIVE mined to at least 10 percent, NEUTRAL to at least 10 percent
□	200-record false-NEGATIVE audit of existing labels completed
□	Dev and test splits human-verified gold; kappa at least 0.6
□	Episode-based split with no leakage; at least 30 test instances per class
□	Dataset frozen and versioned (v3)
□	PhoBERT fine-tuned with diacritic dropout; per-class F1 on gold test
□	Confidence calibrated; ONNX export at or under 50 ms p95
□	Decision layer routing rules implemented and unit-tested
□	Geo-resolution with diacritic normalization, fuzzy matching, provenance flags
□	Event lifecycle: create, resolve, close, cooldown, latency tracking
□	LLM escalation with same prompt, cost cap, and grounding gate
□	Episode replay benchmark run; precision, recall, latency, scalability curve
□	Benchmark table with regex, fine-tuned, architecture, and LLM rows
□	Grounding-provenance statistics and with-or-without-list ablation

Done	Checklist item
□	Undocumented-detection analysis with cross-verified counts
□	Funnel table with posts per day and platform split recorded
□	All seven paper figures and tables prepared
□	Datasheet, ethics statement, license, GitHub, HuggingFace, Zenodo DOI
□	Paper assembled per the nine-section skeleton
□	ICWSM Round 3 submitted by January 15, 2027
□	Defense scripts rehearsed (four challenges of Section 6.3)

Appendix B: Citation Anchors

These are the verified anchors from the research rounds, with the specific figure or claim each one supports. All were checked for open access during the link-verification round; the arXiv, ACL Anthology, and CC-licensed sources are the citable forms. Where a number below appears in the argument text of this document, this table is its provenance.

Table 20: Citation anchors and the claims they support

Anchor	Use in this project
de Bruijn et al. 2019, Scientific Data (88M tweets, 10,000+ events, about 90 percent precision)	Scale-claim precedent; dual historic-real-time evaluation pattern; burst detection parameters
Gu et al. 2026, Nature (92.98M Weibo posts)	Undocumented-flood detection precedent for the gap-finding headline
Kryvasheyeu et al. 2015, PNAS (50M tweets, Sandy)	Single-event mega-corpus precedent; damage correlation approach
Huang et al. 2022, EDEE (3,170 events, 93.56 percent clustering)	Operational system precedent; alert latency distribution reporting
Sakaki et al. 2010, WWW (96 percent detection)	Feasibility-proof pattern for social sensing; small-n output precedent
Earle et al. 2011, USGS (48 events, 2 false triggers)	Small-n credible output precedent
PetaJakarta (1,119 reports per monsoon)	One-city social sensing precedent
CrisisMMD, Alam et al. 2018, AAAI (41K tweets, 2K images)	Small-corpus dataset paper precedent; frame severity label design for Tier 2
DisasterMM, MediaEval 2022	Flood relevance and location extraction benchmark; text-image misalignment evidence

Anchor	Use in this project
Gilardi et al. 2023, PNAS (ChatGPT outperforms crowd workers)	LLM-as-annotator methodology and distillation cover
Zhang et al. 2023, Limits of ChatGPT; Ziems et al. 2024, Computational Linguistics	LLM weakness at structured span extraction; benchmark comparison justification
PASTE (EMNLP 2020); STAGE (AAAI 2022); EMC-GCN (2022); boundary-driven table-filling (2025)	AST architecture families for the benchmark; span-extraction comparability rationale
SemEval-2014 Task 4; SemEval-2016 Task 5	Polarity definition lineage; aspect-category dimension precedent
VLSP 2018 ABSA shared task	Vietnamese ABSA anchor; related-work positioning
PhoWhisper, VinAI (844 hours Vietnamese)	ASR plan for the Tier 1 text merge; WER audit requirement
IRD Dataverse, Historical database of urban flooding in HCMC (425 observations, 2002-2026, DOI 10.23708/8Y16HU)	Long-run ground truth for spatial recall validation
Official 122 flood hotspots (83 rain, 39 tide; October 2026 press publication)	Gazetteer source, soft prior, and one of three evaluation references
DataReportal Digital 2025 Vietnam (76.2M social identities; Facebook 77.6 percent; TikTok 40.9 percent)	Supply-side sensor-network argument